A FINITE FREE EXTENSION IN WHICH A MONIC POLYNOMIAL ACQUIRES A ROOT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A monic polynomial over a ring is divisible by the linear factor corresponding to each of its roots. the statement counted by the sibling card S1-algebra-have-one-root at 4551-4565, cited at proof lines 4579-4580. Produces the quotient polynomial Q in the factorisation.
- The quotient of a polynomial ring by a monic polynomial is a finite free module over the base ring. used at proof lines 4576-4577. Gives finiteness and freeness of the constructed extension.

Source reference. Original lemma 0530: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6588–6588. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 052Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6587–6587. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $P(T)$ be a monic polynomial with coefficients in R. There exists a finite free ring map $R \to R'$ such that $P(T) = (T - \alpha)Q(T)$ for some $\alpha \in R'$ and some monic polynomial $Q(T) \in R'[T]$.*

Proof. Write $P(T) = T^d + a_1 T^{d-1} + \ldots + a_0$. Set $R' = R[x]/(x^d + a_1 x^{d-1} + \ldots + a_0)$. Set α equal to the congruence class of x. Then it is clear that $P(\alpha) = 0$. Thus we win by Lemma 052Z. □

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